BACKUP

Backup slides for the questions

Not part of the 10-minute talk – click a section to jump there

A **Business model and environment**
DESTEP · Porter's five forces · VRIO and SWOT
Slides 14–16 ▸

B **Accounting analysis**
Red flags · room for judgement · adjustments · reliability of ratios
Slides 17–20 ▸

C **Financial analysis**
Growth and costs · DuPont · liquidity and benchmark · cash flow
Slides 21–24 ▸

D **Recommendation, scenarios and risks**
Why 3.9 % is a trigger · scenarios · risks and drop trigger
Slides 25–27 ▸

E **Likely questions and answers**
Six short answers for the Q&A
Slides 28–29 ▸

HOW TO USE

Each backup slide answers one likely question in depth.

Every slide links back here with ◂ **INDEX**.

All figures come from the report and the Excel workbook.

References: slide 30 ▸

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Environment: a modest recovery, but costs rise first

DESTEP scan

Economic

European construction equipment market +4.6 % in 2025, +2 to 2.5 % expected for 2026 – while Liebherr's construction and mining revenue fell 5.5 %.

Political

US tariffs cut Liebherr's North American crane revenue in 2025. Caterpillar expected a tariff impact of $2,600 m in 2026.

Social

Construction had 792 fatal accidents at work in the EU in 2023, 24.0 % of the total. Safety becomes a purchase criterion.

Technological

Autonomous and electrified machines. EU Regulation 2023/1230 covers autonomous functions from 20 January 2027.

Ecological

EU emission limits for non-road machinery and the target of at least 55 % net emission cuts by 2030 make low-emission machines a purchase criterion.

Demographic

Shortages of skilled workers and machine operators push costs up. Personnel expenses reached 28.6 % of revenue in 2025.

Source: Report §3.1.1; CECE (2026), Eurostat (2025), European Union (2023), European Commission (n.d.-a, n.d.-b), Liebherr-International AG (2026), Caterpillar Inc. (2026)

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Competitive forces squeeze the margin

Porter's five forces

New entrants	Moderate, rising	Chinese producers enter with cheaper, simplified machines. Mobile and crawler crane revenue fell 6.8 % in 2025.
Supplier power	Moderate, rising	Cost of materials is 52.7 % of revenue. Electrification shifts power to battery and electronics suppliers.
Buyer power	High, stable	Receivables rose 9.1 % against revenue growth of 1.0 % in 2025. About half of fleet owners may reconsider their brands by 2028.
Substitutes	Low to moderate, rising	Renting and automation change how much equipment is needed. The rental fleet took €305 m, 28.8 % of 2025 investment.
Rivalry	High, rising	The fixed cost of staying in the market rises. The operating margin fell from 4.6 % in 2024 to 2.8 % in 2025.

INVESTOR LENS **Limited industry profit potential – consistent with the fall to 2.8 %**

Source: Report §3.1.2, Table 1; ratings are the judgement of the report

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Hard-to-copy resources – locked in the asset base

VRIO and SWOT

VRIO: what is hard to copy

- **Sustained advantage:** in-house key technologies (40+ production sites in 16 countries), application engineering (R&D €708 m, 4.8 % of revenue), family ownership with permanent capital (equity ratio 53.5 %)
- **Parity only:** wholly-owned sales and service network, decentralised divisional structure
- **Temporary to sustained:** internal capital market

Strengths

Vertical integration · equity ratio 53.5 % · 13 segments

Weaknesses

Margin 2.8 % vs. 17.2 % at Caterpillar · operating cash flow €1,171 m → €530 m · no segment disclosure

Opportunities

Low-emission and automated machines · European market +4.6 % in 2025

Threats

Low-price competitors from China · US tariffs · transition costs for new drives and autonomy

Source: Report §3.2 and §3.4, Tables E1 and D1

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Red-flag screen: reasons to ask, not proof

Nine tests over four years

Four red flags

- **Provision reversals:** €143 m in 2025, 34.6 % of the operating result if credited within it
- **Receivables vs. revenue:** receivables +9.1 % against revenue +1.0 % in 2025
- **Profit vs. tax:** effective tax rate 99.2 %, 45.1 %, 58.2 % and 37.8 % (2022–2025), no reconciliation
- **Impairment:** €55 m in 2023, six times the level of the other years

Clean or not testable

- **Clean:** no off-balance-sheet arrangements disclosed · same auditor (Ernst & Young), no qualified opinion · no change of policy or estimate
- **Not testable:** related party transactions (no note in four years) · large year-end adjustments (no interim figures)

4
flags raised

3
clean

2
not testable

Source: Report §4.4, Table H1; financial statements 2022–2025

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Where management has room for judgement

Rated by the judgement the standard allows and the materiality of the amount

Highest flexibility

- Inventories: €6,341 m, 36.3 % of total assets, allowance not disclosed
- Provisions: €864 m, of which €530 m warranty
- Impairment test
- Deferred tax assets: €285 m

Also material

- Own work capitalised: €317 m, 77 % of the operating result
- PP&E: €4,304 m, machinery written off over 5 to 21 years
- Revenue timing: advance payments €896 m, contract liabilities €289 m

Lowest flexibility

- Borrowing costs
- Lessee accounting
- Translation of foreign currency
- The standard prescribes the mechanics – little choice for management

Source: Report §4.2–4.3, Table G2 (30 treatments, nine of them key)

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22 findings: what we could adjust and what we could not

Short list of the accounting analysis

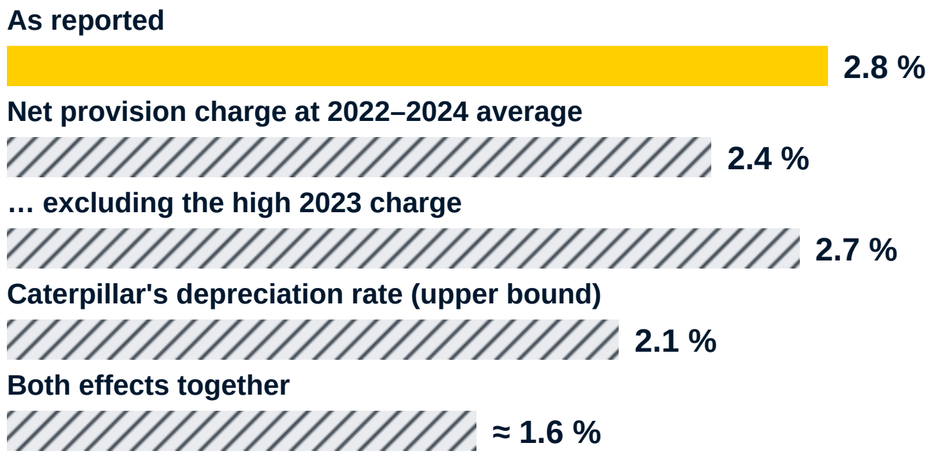
7 **not adjustable**
Red flags and disclosure gaps, e.g. provision reversals of €143 m, missing related party note

5 **reversed in the workbook**
e.g. Caterpillar LIFO → FIFO: equity +$4,305 m, operating profit +$441 m; goodwill impairment 2022

10 **not reversible**
Disclosure too thin, e.g. undefined normal capacity, own work capitalised €317 m

ILLUSTRATIVE SENSITIVITY · NOT BOOKED

Liebherr operating margin 2025



The quantified effects point to a weaker result – the weakness is, if anything, understated.

Source: Report chapter 4, Tables H2, H3 and J13; Excel workbook, sheets C1 and D9

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How far the key ratios can be trusted

After the accounting analysis the ratios fall into three groups

Reliable

- Operating margin trend
- Inventory and payable days
- Current ratio 2.31, quick ratio 1.16
- Cash conversion 53.0 % and free cash flow −€176 m

Usable with bias

- Margin level 2.8 % – scenarios 1.6 % to 2.8 %
- Days sales outstanding 73.8
- Equity ratio 53.5 % (50.4 % at Caterpillar's depreciation rate)
- Fixed asset turnover 3.49

Not usable

- Net margin 1.8 % and return on equity 3.0 % – tax rate 37.8 % to 99.2 %
- Interest cover
- Comparisons with Caterpillar built on equity

Source: Report §6.3, Table J13; 2025 figures; Excel workbook, sheets D3 and D9

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Growth and cost structure 2022–2025

Operating margin per year, with revenue and operating profit

2022	2023	2024	2025
3.6 %	**4.4 %**	**4.6 %**	**2.8 %**
Revenue €12,589 m	Revenue €14,042 m	Revenue €14,622 m	Revenue €14,772 m
Operating profit €454 m	Operating profit €619 m	Operating profit €674 m	Operating profit €413 m

Where growth stalled

- Revenue growth 11.5 % (2023) → 4.1 % (2024) → 1.0 % (2025)
- Construction and mining −5.5 % to €9,345 m
- Other segments +14.7 % to €5,427 m, components +35.5 %

Where the margin went

- Personnel expense +5.3 % to €4,226 m vs. revenue +1.0 %
- Personnel share of revenue 26.4 % (2023) → 28.6 % (2025)
- Other operating items cost another 1.25 pp
- Each margin point = €148 m operating profit

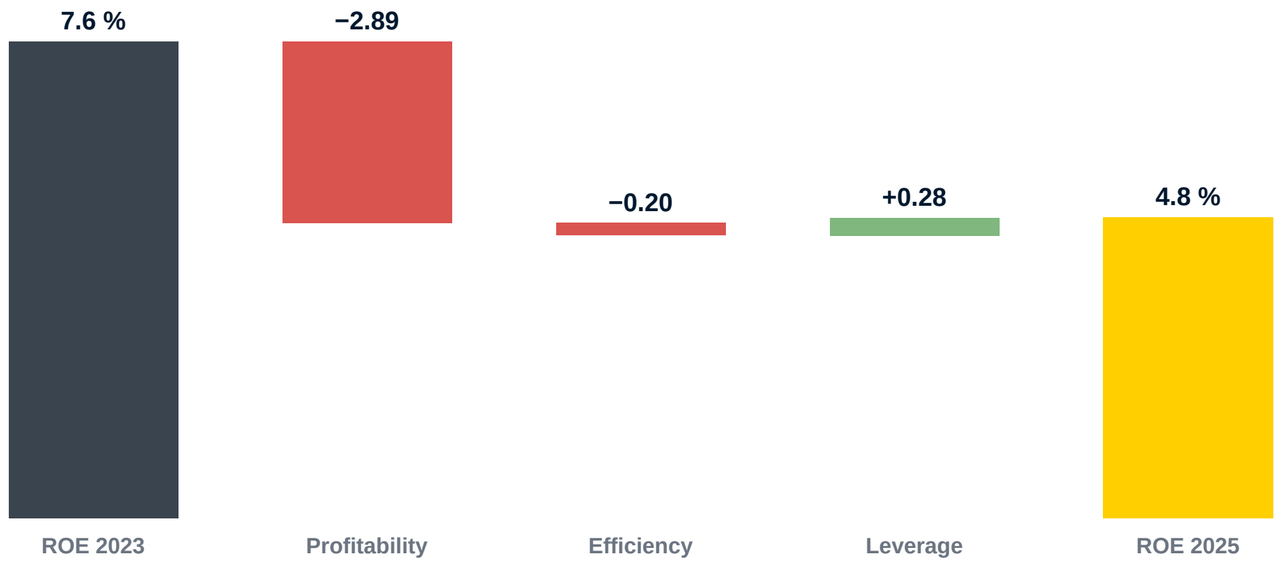
Source: Report §5.2; Excel workbook, sheets D1 and D6

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DuPont: the margin, not leverage, drives returns down

Pre-tax return on equity 2023 → 2025

Contributions to the change in pre-tax ROE, percentage points



pre-tax return on equity · contributions in pp

What it means

- **Profitability −2.89 pp:** lower pre-tax margin
- **Efficiency −0.20 pp:** slightly lower asset turnover
- **Leverage +0.28 pp:** more debt even helped a little
- After tax, ROE was 3.0 % vs. 16.4 % in the US industry – but the tax charge is volatile, so the trend is read before tax

Source: Report §5.3, Figure 2a, Table J4; Excel workbook, sheets D4 and D6 (logarithmic split)

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Strong solvency, far weaker returns

Liquidity, solvency and benchmark 2025

Balance sheet strength

- Equity ratio 53.5 %; bank and lease liabilities 23.7 % of capital (industry 39.3 %)
- Cash and current financial assets €3,147 m cover current debt of €1,215 m 2.6×
- Current ratio 2.31 vs. 1.55 at Caterpillar
- Net financial position €523 m (2022) → net debt €85 m (2023) → €249 m (2025)
- Debt / EBITDA 2.14 → 2.90; interest cover 11.07 (2022) → 3.90 (2025)

2025	LIEBHERR	US INDUSTRY	CATERPILLAR
Operating margin	**2.8 %**	15.9 %	17.2 %
EBITDA margin	**6.8 %**	19.6 %	19.8 %
Return on equity	**3.0 %**	16.4 %	38.1 %*
Inventories, % of revenue	**42.9 %**	16.6 %	33.2 %
Debt / EBITDA	**2.90**	2.30	3.52

* Not comparable – Caterpillar's equity is reduced by treasury shares. Caterpillar on the FIFO basis.

Source: Report §5.1 and §5.5, Table J14; Damodaran (2026b, 2026c, 2026d, 2026e), US machinery industry, January 2026; Excel workbook, sheets D3, D7 and D10

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Liebherr invests ahead of its cash

Cash flow and capital allocation

190.2 %
Capex in % of operating cash flow, 2022–2025

–€1,151 m
Free cash flow in total, 2022–2025

28.4 %
of EBITDA became operating cash flow, 2022–2025 (53.0 % in 2025)

€530 m
Operating cash flow 2025, down from €1,171 m (2024)

Capex above operating cash flow

- Capex and R&D 9.6 % of revenue vs. 7.4 % at Caterpillar – at a margin of 2.8 % vs. 17.2 %
- Shortfall after distributions €388 m (2023–2025)
- Bank and lease liabilities +€562 m (2022–2025)

Caterpillar returns cash

- Free cash flow $7,453 m in 2025, 11.0 % of revenue
- 73.1 % of operating cash flow returned to shareholders (2022–2025)
- Liebherr distributed €56 m in 2025 – little current income for an investor

Source: Report §5.6, Figure 4; Excel workbook, sheets D5, D7 and D10

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Why 3.9 % is only a trigger, not a target

Condition 1 of the reassessment

3.9 %
Average operating margin 2022–2025 (3.86 %)

+€157 m
Operating profit at 2025 revenue (€14,772 m)

3.5 %
ROCE before tax 2025 vs. 7.70 % after-tax benchmark

6–8 %
Operating margin needed to earn the benchmark

What the trigger means

- A return to 3.9 % is the first realistic sign of recovery – Liebherr has earned it on average
- It is a test, not a target return
- Each margin point is worth €148 m

Why it is not enough

- Even in 2024, at a 4.6 % margin, ROCE was only 5.9 % before tax – below the 7.70 % benchmark
- The benchmark is the US machinery industry, not Liebherr's own cost of capital
- A margin recovery alone does not prove an adequate return

Source: Report §5.3 and §5.7, chapter 8; Excel workbook, sheets D3 and D10; Damodaran (2026a)

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What the scenarios are worth

Operating profit at 2025 revenue (€14,772 m) – illustrations, not forecasts

Margin scenarios

2025 actual · 2.80 % margin



€413 m

Four-year average · 3.86 % margin



€570 m **+€157 m**

Best year 2024 · 4.61 % margin



€681 m **+€268 m**

US industry average · 15.86 % margin

€2,343 m **+€1,930 m**

The industry figure shows the size of the gap and is not a target.

Cash scenarios (2025)

- Free cash flow −€176 m: the net financial position of €249 m covers it 1.4 years
- After distributions (−€232 m): 1.1 years
- Capex at the level of depreciation (€587 m): +€119 m, free cash flow still −€57 m
- Cash conversion cycle 10 days shorter: €405 m released once

Source: Report Table J15; Excel workbook, sheet D10. Margins Liebherr has earned or the industry average – nothing is assumed

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Risks, limits and when to drop the case

What could still go wrong – and the alternative for a cautious investor

Demand

Construction and mining revenue −5.5 % in 2025; no significant upswing expected before 2027

Costs

US tariffs and the cost of autonomous and electrified machines press on the margin

Information

The largest cost increase of 2025 cannot be explained from the summary statements

Valuation

Not assessed in the report – price and contract rights decide the case

Alternative for a risk-averse investor

Equity is unsuitable: low payout, shares not traded. A loan may fit better, as equity of 53.5 % limits the default risk – but pricing, maturity, collateral and covenants are not disclosed.

Drop trigger

If the margin stays near 2.8 % in 2026 **and** the net financial position turns negative, the investor should drop the case rather than wait for 2027. A recovery should not rest on large provision releases.

Source: Report §5.5, §5.7, §6.1 and chapter 8

Why not invest now?

Likely questions (1/2)

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Q Why not invest now?

Neither financial condition was met in 2025: the margin of 2.8 % is below its four-year average of 3.9 %, and free cash flow was −€176 m. The answer is **not yet**, rather than no.

Q Why is 3.9 % the first test?

It is the average Liebherr earned from 2022 to 2025 – a realistic first sign of recovery. At 2025 revenue it adds €157 m of operating profit. A trigger, not a target return.

Q Is the LIFO to FIFO adjustment fair?

IAS 2 does not permit LIFO, so FIFO puts both groups on one basis. Caterpillar's LIFO reserve of $4,305 m lifts its margin only from 16.50 % to 17.15 % – the gap stays large.

Source: Report chapters 4 to 8; Excel workbook, sheets C1, D3 and D10

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What the balance sheet cannot fix

Likely questions (2/2)

Q What do the red flags mean?

They are reasons to ask, not proof of manipulation. Four flags are raised, three tests are clean and two cannot be run on the published statements – that is why information rights come first.

Q Why is ROCE so low?

ROCE is 3.5 % before tax against a 7.70 % after-tax benchmark: the margin is only 2.8 %, and inventories of 42.9 % of revenue and a cash cycle 77 days longer than Caterpillar's tie up capital.

Q Why is a strong balance sheet not a return?

An equity ratio of 53.5 % and a net financial position of €249 m limit the risk, but do not earn the cost of capital. They would cover a shortfall like 2025 (−€232 m) for about 1.1 years.

Source: Report chapters 4 to 8; Excel workbook, sheets C1, D3 and D10

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References

Sources cited in the presentation, including the backup slides (APA 7th edition)

Caterpillar Inc. (2023). *Annual report on Form 10-K for the fiscal year ended December 31, 2022*. U.S. Securities and Exchange Commission.

Caterpillar Inc. (2026). *Annual report 2025* [Including the annual report on Form 10-K for the fiscal year ended December 31, 2025]. U.S. Securities and Exchange Commission.

Committee for European Construction Equipment. (2026, February 26). *European construction equipment sector sees moderate recovery in 2025 with 4.6% growth in sales*.

Damodaran, A. (2026a). *Cost of equity and capital (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026b). *Debt fundamentals by sector (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026c). *Margins by sector (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026d). *Return on equity by sector (US)* [Data set]. NYU Stern School of Business.

Damodaran, A. (2026e). *Working capital ratios by sector (US)* [Data set]. NYU Stern School of Business.

EUROCONSTRUCT. (2026, June 5). *Press release – 101st EUROCONSTRUCT conference* [Press release].

European Central Bank. (2026, September 10). *Monetary policy decisions* [Press release].

European Commission. (n.d.-a). *Emissions from non-road mobile machinery*.

European Commission. (n.d.-b). *European climate law*.

European Union. (2023). Regulation (EU) 2023/1230 of the European Parliament and of the Council of 14 June 2023 on machinery and repealing Directive 2006/42/EC of the European Parliament and of the Council and Council Directive 73/361/EEC. *Official Journal of the European Union*, L 165, 1–102.

Eurostat. (2025). *Accidents at work – statistics by economic activity*.

IFRS Foundation. (n.d.). *IFRS Accounting Standards Navigator*.

Liebherr-International AG. (2023). *Annual report 2022*.

Liebherr-International AG. (2024). *Annual report 2023*.

Liebherr-International AG. (2025). *Annual report 2024*.

Liebherr-International AG. (2026). *Annual report 2025*.

Own work: Beermann, S., Fliegel, A., Narkiewicz, Z., Hoffmann, P., & Gilmore Islam, Z. (2026). *Liebherr Group financial report: Investment analysis 2022–2025* and the Excel workbook *Liebherr–Caterpillar restructured financial statements*. Fontys University of Applied Sciences, Venlo. · Images: Liebherr-International AG. · Full entries with URLs: report, References.